

Statistics of Paramagnetism

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N magnetic dipoles interacting with a magnetic $\vec{H}$. Magnetic moment of dipoles $\vec{\mu}$.
We consider in the simplest case that dipoles are fixed and localised.

Completely localised dipoles. No translation only rotation.

Classical treatment

$$E = \sum_{i=1}^N E_i = - \sum_{i=1}^N \vec{\mu} \vec{H} = - \sum_{i=1}^N \mu H \cos \theta_i \quad (1)$$

as all dipoles have the same $|\vec{\mu}| = \mu$.

Thermodynamic limit

$$Z = \prod_{i=1}^N \zeta_i = \zeta^N \quad (2)$$

because there is no interaction between dipoles.

We have the canonical partition function for one dipole

$$\zeta = \zeta(T, V, 1) = \sum_{\theta, \phi} e^{+\beta \vec{\mu} H \cos \theta} = \int d\theta d\phi \sin \theta e^{-\beta \vec{\mu} H \cos \theta} \quad (3)$$

$$= \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta e^{+\beta \vec{\mu} H \cos \theta} \quad (4)$$

$$= (2\pi) \int_{-1}^1 d(\cos \theta) e^{+\beta \vec{\mu} H \cos \theta} \quad (5)$$

if we let $x = \beta \vec{\mu} H \cos \theta$

$$= \frac{2\pi}{\beta \vec{\mu} H} \int_{-\beta \vec{\mu} H}^{\beta \vec{\mu} H} dx e^{-x} = \dots \quad (6)$$

$$\implies \zeta(\beta, V, N=1) = \frac{4\pi \sinh(\beta \vec{\mu} H)}{\beta \vec{\mu} H} \quad (7)$$

$$Z = \zeta^N = \left(\frac{4\pi \sinh(\beta \vec{\mu} H)}{\beta \vec{\mu} H} \right)^N \quad (8)$$

$$\mathcal{F} = -kT \ln Z = -kTN \ln \zeta \quad (9)$$

We can determine the thermo behavior of the system.

$\langle \vec{\mu} \cos \theta \rangle ==$ average dipole moment of each dipole.

$N \langle \vec{\mu} \cos \theta \rangle == \dots$ of the total system.

We can find everything from our free energy in the classical limit.

$$\langle \vec{\mu} \cos \theta \rangle = \frac{\int \int d\phi d\theta \sin \theta \vec{\mu} \cos \theta e^{-\beta \vec{\mu} H \cos \theta}}{\int \int d\phi d\theta \sin \theta e^{-\beta \vec{\mu} H \cos \theta}} \quad (10)$$

$$= \frac{1}{\zeta \beta} \left(\frac{\partial}{\partial H} \zeta \right)_{\beta, V} = \frac{1}{\beta} \left(\frac{\partial \ln \zeta}{\partial H} \right)_{\beta, V} \quad (11)$$

$$\implies -\frac{1}{N} \left(\frac{\partial \mathcal{F}}{\partial H} \right)_{\beta, V, N} == \langle M \rangle \quad (12)$$

χ_t is the isothermal magnetic susceptibility $\chi_T = \left(\frac{\partial \langle M \rangle}{\partial H} \right)_{\beta, V, N}$ We can evaluate

$$\langle M \rangle = \dots = -\frac{1}{\beta H} + \vec{\mu} \coth(\beta \vec{\mu} H) = \vec{\mu} \left(-\frac{1}{\beta \mu H} + \coth(\beta \vec{\mu} H) \right) \quad (13)$$

We can let $x = \beta \vec{\mu} H$ and define the function

$$\mathcal{L}(x) = \left(-\frac{1}{\beta \mu H} + \coth(\beta \vec{\mu} H) \right) \quad (14)$$

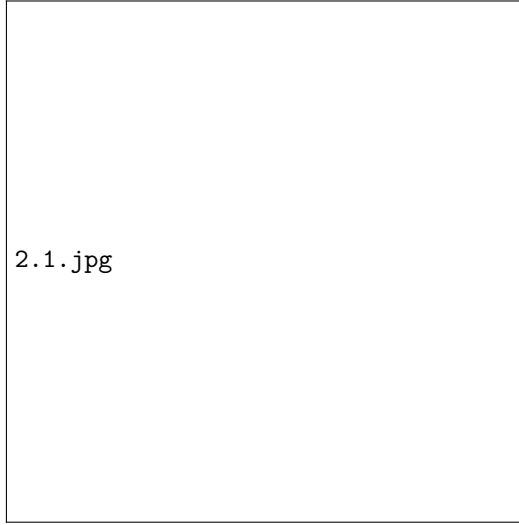


Figure 1: pg 84 beale

$$\hat{H} = -\vec{\mu} \vec{H} \quad (15)$$

quantisation

$$\vec{\mu} = g \left(\frac{q}{2mc} \right) () \quad (16)$$

where g is the gyro magnetic ratio.
either $\vec{L}$, $\vec{S}$ or combination $\vec{J}$.

$$E = -g \vec{\mu}_0 m H \quad (17)$$

m is the quantum number $m = -J, -J+1, \dots, J-1, J$. Notice there are $2J+1$ allowed values.

$$\vec{\mu}_0 \frac{q \hbar}{2mc} \quad (18)$$

$$\zeta(\beta, V, H, N=1) = \sum_{m=-J}^J e^{(\beta g \vec{\mu}_0 H) m} \quad (19)$$

let $x = \beta g \vec{\mu}_0 H J$

$$\zeta = e^{-x} \left(\frac{e^{\frac{2J+1}{J} x} - 1}{e^{\frac{x}{J}} - 1} \right) \quad (20)$$

$$\zeta = \dots = \frac{\sinh \frac{2J+1}{J} x}{\sin \frac{x}{2J}} \quad (21)$$

$$Z = \zeta^N, \mathcal{F} = -kT \ln Z \quad (22)$$

$$\langle M \rangle = - \left(\frac{\partial \ln \zeta}{\partial H} \right)_{\beta, V} \quad (23)$$

$$\chi_T = \left(\frac{\partial \langle M \rangle}{\partial H} \right)_{\beta, V} \quad (24)$$